Characters between the puzzles

A small flat 2D cast for short story consequences after answers. The active exercise gives its space to the math task.



Concept board only. Remove residual shading and simplify details in production. Written shapes and palette tokens take precedence.

Confirmed direction

Flat 2D characters. Production by an AI agent. Decorative scenes appear between exercises and at completion. Mathematical diagrams stay in the exercise.

Proposed scope

Four recurring characters. Start with Cat and Boba, then build one picnic sequence. Alder and Ada follow only if requested. Appearance, personalities, timings, and quantities remain proposals.

This package is a plan and visual reference. Production SVGs, animation frames, and app integration have not been created.

Four characters, distinct behavior

Player cat

Curious and persistent. Helps the other character and shares the result with the player.

Orange tabby; cream forehead blaze, muzzle, chest and paws; triangle ears; hooked striped tail; small gold tag. Keep four-legged anatomy. The displayed name uses the saved pet name.

Motion: Head tilt, then a forepaw gesture, with the tail following slightly later.

Boba the rabbit

Eager and practical. Impatient with a wobbly object, never with the child.

Cream pear-shaped body, long pink-inner ears with one bent tip, teal cap, broad back feet and a small round tail.

Motion: Ears prepare the action. A small wave or low hop ends with both feet settled.

Alder the beaver

Methodical and quietly pleased when parts fit. A collaborator, not an examiner.

Broad low brown body, rounded-square head, small paired incisors, paddle tail and golden apron with one pocket. Limit tail cross marks to three.

Motion: Grounded nod, deliberate paw movement and a restrained two-paw celebration.

Ada the owl

Observant and imaginative. Enjoys discovering things and can make harmless mistakes.

Lilac body, plum wings and ear tufts, pale angular face mask, almond eyes with amber accents, three lower feather forms. No glasses or clothing.

Motion: Head tilt, pleased blink and a short wing flourish. No spinning head or full flight cycle.

Shared drawing rules

Closed shapes, solid fills and rounded charcoal outlines. No textures or gradients. At a 512 px source size, use 5 px outer strokes and 3 px interior marks. Check expression clarity at 96 px character height. Exact color tokens are in asset-plan.json.

Story follows the answer

Moment	What the child sees
Solving	Instruction, math board, answers, help and Check answer. No decorative illustration or idle character.
Correct stage	Explanation and one short scene consequence in the existing result screen. Continue is available immediately.
Wrong answer	The math remains visible with specific retry guidance. No large sad scene or punishment performance.
Final stage	One longer payoff in the result flow. Hold the finished scene. Do not add another celebration page.

Pilot story: prepare a picnic

Cat and Boba use one basket, one lunch bundle and one blanket. The sequence communicates progress; it contains no answer clues, numbers or baked text.

Checkpoint	Transition to the next state
0: Ready	Closed empty basket, lunch bundle outside, folded blanket. Boba opens the basket while Cat looks. 1 second.
1: Open	Basket open, bundle outside. Cat slides the bundle in; Boba steadies the handle and closes the lid. 1 second.
2: Packed	Closed packed basket, folded blanket. Boba unfolds the blanket and places the basket beside it. Both settle. 2 seconds.
3: Complete	Unfolded blanket, packed basket, two pleased characters. Hold this still after the ending.

Later stories

Alder: align bridge supports, fit one deck piece, then let Cat cross safely. **Ada:** align loose pages, restore the cover, then open the book together. Keep both cameras fixed.

Integration must replace the selected mission title and all stage copy coherently in English and Latvian. Do not combine the old branch-rescue title with balloon-launch or new picnic scenes. Preserve puzzle and save IDs.

A small reusable motion library

Clip	Length and export	Meaning
greet	1 second / 8 frames	Optional mission entry or a new character introduction.
acknowledge	1 second / 8 frames	A restrained stage success.
celebrate	2 seconds / 16 frames	The mission is complete.

Each character receives these three logical clips. Cat and Boba supply six for the pilot. Their scene interactions use the same rigs with scene-specific choreography.

Pose timing

One-second action	Two-second action
0 ms: establish the starting pose. 125-250 ms: prepare. 250-500 ms: main gesture. 500-750 ms: settle. 750 ms onward: hold.	0-375 ms: establish and prepare. 375-1125 ms: main action. 1125-1500 ms: settle. 1500 ms onward: hold.

These are proposed project timings, not measurements of Duolingo. Keep head turns near 12 degrees and body squash/stretch around 5 percent. Use clear preparation and landing for any hop. Motion must read with sound off.

Playback contract

Play once, then hold. The final frame matches the destination still. Continue never waits for the clip. Reduce Motion shows the destination still immediately with the same explanation and action.

Pause when backgrounded. Returning can resume or show the destination still. Missing media falls back to text and a known still. The renderer never awards rewards, spends lives or saves a stage.

Stable source art

Build real layered SVG masters and animate their named parts with deterministic keyframes. Record each pivot and hierarchy. Reuse source geometry across every pose and frame. Never generate each animation frame as a new AI image.

Do not add continuous idles, lip sync, new audio or a new animation engine for this pilot. Keep one focal scene in each result and suppress the duplicate companion portrait there.

Production batches and delivery

Batch	Deliverable
1: Masters	Cat and Boba: 2 character SVG masters, 2 model sheets as SVG and PNG, rig metadata, local preview. Front, three-quarter, side, silhouette and 6 face states per character.
2: Character motion	6 atlases, 6 final stills, 64 PNG frames, deterministic timelines and replayable preview. Use the same two character masters.
3: Picnic kit	4 checkpoint stills, 3 composite atlases, 32 PNG frames, scene/timeline source and a preview of all transitions.
4: Native pilot	Integrate one selected boss, update localized story copy and check navigation, accessibility, checkpoints and rewards.
5: Expansion	Only when requested: Alder and Ada add 6 character clips; bridge and book add 2 scene kits.

Export profile

Item	Requirement
Character source	512 x 512 SVG. Default three-quarter view facing viewer right. Floor anchor [256, 464]. Named groups and documented pivots.
Scene source	1024 x 640. Final frames 512 x 320 RGBA PNG. 24 px safe inset in the final frame, including every moving part.
Runtime atlases	8 fps, 125 ms per frame, 4 columns, row-major order. 8 frames: 2048 x 640. 16 frames: 2048 x 1280. Transparent WebP.
Media target	Provisional 2 MiB compressed pilot budget. A long atlas decodes to about 10 MiB; load only the active clip and measure on native devices.

Pilot totals: 2 characters, 6 generic clips, 3 composite scene clips, 4 scene states and 96 delivery-frame PNGs. Sheets and stills are additional files. Orange is the pilot coat. Generate grey and white derivatives from shared geometry and color tokens before a release that represents the player's pet.

Start the next agent here

Give the agent the entire package and the First run prompt from AGENT_PROMPTS.md. It begins with Cat and Boba model sheets and editable masters only. Each later prompt is a separate bounded task.

Files to read

File	Purpose
START_HERE.md	Detailed creative brief, behavior, scene choreography, production rules and repository map.
asset-plan.json	IDs, exact palettes, dimensions, frame counts, planned paths and playback contract.
AGENT_PROMPTS.md	First run, animation, scene, integration and expansion tasks. Required status report for every run.
concepts/	Proposed lineup and generation notes. Written requirements override residual shading or extra detail.

Review each batch before expanding it

Check silhouettes at 96 px. Inspect every frame for clipped ears or tails, changing markings, sliding feet and floating props. Check cream and charcoal backgrounds for alpha fringes. Match every scene endpoint to its checkpoint still. Preview at 320 px scene width.

Native integration adds checks for early Continue, repeated taps, mistakes, resume, replay, final completion, Reduce Motion and backgrounding. Static previews do not prove device performance. Preserve the 3D home and existing player data.

What the research contributes

Duolingo documents character reactions and mid-lesson interludes, simple readable shapes, and state-driven animation. It also uses characters inside some exercises. This project deliberately adopts a stricter puzzle-only active screen. The cast and production timings here are original proposals.

Character reactions and interludes
https://blog.duolingo.com/building-character/

Shape language and readability
https://blog.duolingo.com/shape-language-duolingos-art-style/

State-driven animation
https://blog.duolingo.com/world-character-visemes/

Mathematical visuals
https://blog.duolingo.com/developing-math/